Supplementary Material

Machine learning in battery management systems: a two-tier scoping review of 799 eligible studies with in-depth classification and validation audit

This supplement consolidates the robustness material that supports the main text: the interpretation rules that govern every quantitative claim (S1), the label-noise scenario envelope for the validation counts (S2), the minimum-detectable-effect analysis of the method-family contrast (S3), the construction and limits of the recall audit (S4), and the two abstract-level probes (S5). Every section states the released artifacts it derives from; the complete evidence base and analysis pipeline are deposited at Zenodo, DOI 10.5281/zenodo.23436202, with the GitHub repository of the same name as a development mirror. The main text carries one-sentence summaries of each item with pointers here, so the supplement can be read alongside Sections 2.2, 2.3, and 3.4 of the paper.

S1. Interpretation rules for every quantitative claim in the paper

Five rules govern how the paper's numbers may be read. They are stated here in full because the main text invokes them by number.

Rule 1 (subset-scoped claims). Anything coded from full texts — function, method family, chemistry, data source, validation protocol, baseline screen — describes the 103-study classified subset only and is never generalized to the field. Headlines that read 'the subset's' or 'within the classified subset' invoke this rule.

Rule 2 (pool-scoped claims). Tier-1 counts (799 records) measure a capped 30-query Crossref retrieval, not a census. Per-year counts are observed counts inside that cap; query-theme tag shares are read for ordering, not exact rates; the 2024 window is partially covered (no query pass was dedicated to January–May 2024); and the recall audit of S4 (16–17% capture of the reviews' own cited literature) bounds external validity.

Rule 3 (earliest means earliest in the classified subset). Recall is bounded, so 'first' and 'earliest' claims are subset-first by construction and are phrased accordingly.

Rule 4 (non-significance is not equivalence). The physics-informed versus data-driven contrast is underpowered (minimum detectable effect roughly 30–35 percentage points, S3); 'no detectable difference' is reported as exactly that.

Rule 5 (curation conditions the subset). Tier-2 selection is quota- and citation-ranking-driven (main text, Section 2.3), so distributions inside the 103 partly reflect those choices and are reported as corpus composition.

Invoked at: main text Sections 2.2, 2.3, 2.6, 3.2–3.4, and the Figure 5 caption. Released artifacts: selection.json, screening_log.csv, pool.csv, recall_audit.json.

S2. Label-noise scenario envelope for the validation counts

The strong/weak validation counts of the main text rest on a single full-text coding by the lead author (the reliability study of Section 2.5 measures abstract-level reproducibility, not full-text agreement). To quantify how much those counts depend on the labels, a label-noise scenario analysis perturbs them directly. The analysis is a scenario envelope, not an error model.

Design. In each of 4,000 seeded Monte Carlo draws (seed 7), a label-flip rate is drawn from a Normal distribution centered on an observed coder-versus-label error rate with standard deviation 0.02, truncated at zero; the drawn rate then flips strong/non-strong bucket labels across the 88 primary papers, and both the overall strong share and the physics-versus-other Fisher exact test are recomputed.

Scenario 1 (symmetric). The rate is centered on the coders' overall bucket error rate against the released labels (0.37 and 0.27 for the two coders; mean 0.32), and flips labels in either direction. Symmetric random flipping mechanically attenuates contrasts toward the null.

Scenario 2 (strong-to-weak, worst-case bound). The actually observed coder-label disagreements are directional and concentrated on the strong labels: 16 of the 19 bucket disagreements have the coder assigning weak where the released label says strong (the declared conservative fallback of Section 2.5), 13 of the 30 sampled papers are strong-labelled, and each coder recoded 8 of those 13 weak — a strong-label-specific rate of 0.62 (16 of 26 pooled coder-labels). Scenario 2 therefore flips strong labels to weak only, at a rate drawn from a Normal centered on that strong-label-specific rate. Because the coders worked from abstracts under a declared weak-if-unreported rule, that rate is an upper bound on plausible label error, and Scenario 2 is reported as a worst-case bound, not an estimate.

Results. Under the symmetric scenario the strong-protocol count spans 35–52 of the 88 primary papers (39.8–59.1%) and the family contrast remains non-significant in 94.4% of draws. Under the strong-to-weak worst-case bound the count falls to about 16 of 88 (≈18%) at the observed recode rate, with a scenario envelope of 10–23 (11.4–26.1%), and the contrast remains non-significant in 97.6% of draws. Read together, the two scenarios bracket the single-coder counts around the documented disagreement structure — under either, the physics-versus-other contrast shows no detectable difference and remains underpowered in the sense quantified in S3 — and the worst-case bound says only how low the strong-protocol count could sit if abstract-level recoding were taken at face value, not where it sits.

Released artifacts: sensitivity_label_noise.py and sensitivity_label_noise.json; gold_labels_30.csv (the 30-sample labels); coder_B_30.csv, coder_C_30.csv, agreement_report.csv, dispute_items.csv, and the hash-tracked reliability log under corpus/reliability/.

S3. Minimum-detectable-effect analysis of the method-family contrast

The main-text contrast — 8 of 22 physics-informed/hybrid primary papers with a strong protocol versus 34 of 66 others (protocol-only definition) — is accompanied by a power analysis so that non-significance is not misread as equivalence.

Method. Analytic two-proportion power, validated against Fisher-exact Monte Carlo simulation, is computed for the observed arm sizes and a 51.5% baseline strong-protocol rate. Under the broader definition of stringency (which also counts experimental-hardware demonstrations) the same analysis is repeated for the 13-of-22 versus 49-of-66 contrast.

Results. With 22 versus 66 papers, 80% power at $\alpha = 0.05$ requires a difference of roughly 30–35 percentage points between the families (odds ratio ≤ 0.26 or ≥ 4.0); the observed 15.2-point gap is detectable with only about 20–25% power. The broader-definition contrast is similar. The two arms' Wilson 95% intervals overlap almost completely (19.7–57.0% versus 39.7–63.2%). The honest statement is that no difference is detectable at this scale, and that meaningful differences in either

direction — including a physics prior halving or doubling the odds of a strong protocol — remain compatible with these counts.

Released artifacts: power_analysis.py and power_analysis.json.

S4. Recall-audit construction, yields, and limits

The main text reports that the 1,550-record retrieval contains 16–17% of the in-scope literature cited by this review's own included reviews. This section states how that number is built.

Denominator. The deposited reference lists of the 15 included reviews and the six seed studies — 2,508 references in total — are restricted to references dated 2020 or later (1,367 are dated before 2020 and fall outside the retrieval window by design), then to references that pass the same title gate as the screening pipeline and carry matchable DOIs: 649 counted with multiplicity, 586 unique.

Numerator and result. The 1,550-record retrieval contains 112 of the 649 (17.3% per-source) and 96 of the 586 unique references (16.4%); all 96 are Tier-1 eligible. Per-query yields are design-capped, not saturated: every pass returned its top 20 records only, so the 30 query families are capped at 100 records each; the five passes yielded 3,000 fetched rows, 2,973 after document-type, year, and title-completeness filters, and 1,550 unique records after deduplication.

Limits. The denominator is a citation-conditioned sample of what the reviews happened to cite; the in-scope gate runs on titles; and references without deposited DOIs cannot be matched. The audit is therefore a floor on external validity for theme frequencies, per-family shares, and 'earliest' claims, which is why all such claims in the paper are subset- or pool-scoped (S1, rules 2 and 3). The recall miss documented in main-text Section 8.4 (the TEC 2025 GPT-4 estimator) fits this pattern exactly: it is cited by none of the included reviews either, so no reference-list audit would have surfaced it.

Released artifacts: recall_audit.py, recall_audit.json, and the cached Crossref responses under raw_cache/.

S5. Abstract-level probes

S5.1 Representativeness probe: can the pool carry the validation audit?

Question. Does the curated subset overstate validation evidence relative to the pool it was drawn from — at the one level where the 799-record pool is observable, its indexed abstracts?

Design. A seed-pinned uniform sample of 40 eligible records was drawn (released with its codes); 23 abstracts were retrievable from public indexes (Elsevier-withheld abstracts dominate the 17 missing). Each was coded under a three-value rubric: isolation or deployment protocol stated; evaluation stated without isolation; nothing stated. The pool sample's record mix: 16 primary-ML, 2 model-based, 3 review, 2 other.

Result. None of the 18 primary-looking pool records states an isolation or deployment protocol in its abstract (0 of 18), against 5 of the 38 abstract-screenable subset papers (13%); the difference is not significant (Fisher's exact $p = 0.16$) and both rates are near zero. The finding is negative in a useful way: indexed abstracts almost never carry enough protocol information to code validation strength at pool scale, in either tier, so the 48% strong-protocol share cannot be promoted from a curated-subset statistic to a field estimate without full-text access — and the scarcity of protocol statements in abstracts is itself part of what the reporting checklist of main-text Section 9.9 responds to.

S5.2 Baseline-comparison screen: do ML studies benchmark against model-based methods?

Question. The review had audited validation protocols but not whether ML studies compare against model-based or classical reference methods at all — a Kalman-family filter (EKF/UKF), an observer, an equivalent-circuit or physics-based model, coulomb counting, or a classical regressor such as Gaussian-process regression.

Design. A seventh, lighter classification field (added when the review's own checklist turned on its authors) records, from indexed abstracts, whether such a reference method is named as a comparator: values none, mentioned, and stated same-data comparison. It is coded from abstracts, not full texts; the released rubric (codebook.md) also defines the full-text version of the field for future work.

Result. Of the 38 abstract-screenable primary papers, 32 (84%) name no model-based or classical baseline, 2 name one without stating the comparison, and 4 (11%) state a same-data quantitative comparison — against an SVR, a 0.5C CCCV protocol, an equivalent-circuit/electrochemical model, and a rule-based controller, respectively. These are lower bounds on comparison practice (full texts may compare what abstracts omit) but exact measures of comparison visibility, and visibility is what a reader scanning the literature actually has: at abstract level, the field's answer to 'does the learned estimator beat the model-based baseline on the same data?' is, four times in five, unstated.

Released artifacts: abstract_probe.py, abstract_probe_raw.json, abstract_probe_codes.py, abstract_probe.json; rubric in codebook.md.

All scripts run from the repository root against the released data and reproduce their outputs deterministically (seeded where stochastic). Deposit of record: Zenodo, DOI 10.5281/zenodo.23436202; development mirror: the GitHub repository of the same name.